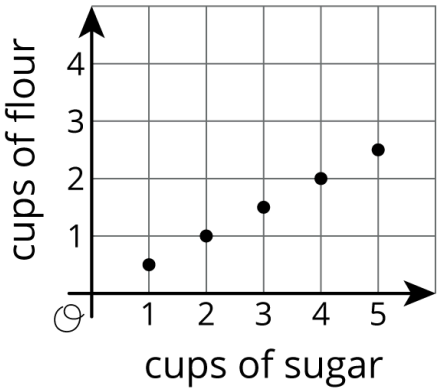
Name _____ Date _____ Period _____

Linear Relationships in Context

Grade 8 · Linear Relationships

1. A restaurant offers delivery for their pizzas. The total cost is a delivery fee added to the price of the pizzas. One customer pays $25 to have 2 pizzas delivered. Another customer pays $58 for 5 pizzas. How many pizzas are delivered to a customer who pays $80?

2. The graph shows the relationship between the number of cups of flour and the number of cups of sugar in Lin's favorite brownie recipe.



The table shows the amounts of flour and sugar needed for Noah's favorite brownie recipe.

cups of sugar	cups of flour
$\frac{3}{2}$	1
3	2
$4\frac{1}{2}$	3

a. Noah and Lin buy a 12-cup bag of sugar and divide it evenly to make their recipes. If they each use all their sugar, how much flour do they each need?

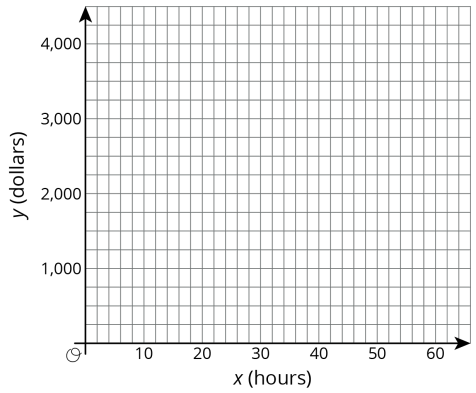
b. Noah and Lin buy a 10-cup bag of flour and divide it evenly to make their recipes. If they each use all their flour, how much sugar do they each need?

3. To paint a house, a painting company charges a flat rate of $500 for supplies, plus $50 for each hour of labor.

a. How much would the painting company charge to paint a house that needs 20 hours of labor? A house that needs 50 hours?

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b. Draw a line representing the relationship between x, the number of hours it takes the painting company to finish the house, and y, the total cost of painting the house. Label the two points from the earlier question on your graph.



c. Find the slope of the line. What is the meaning of the slope in this context?

4. Write an equation for the line that passes through $(2, 5)$ and $(6, 7)$.

5. Tyler and Elena are on the cross country team.

Tyler's distances and times for a training run are shown on the graph.



Elena's distances and times for a training run are given by the equation $y = 8.5x$, where x represents distance in miles and y represents time in minutes.

a. Who ran farther in 10 minutes? How much farther? Explain how you know.

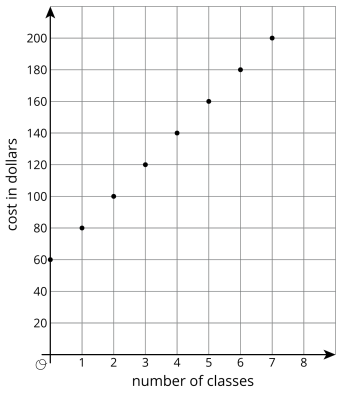
b. Calculate each runner's pace in minutes per mile.

c. Who ran faster during the training run? Explain or show your reasoning.

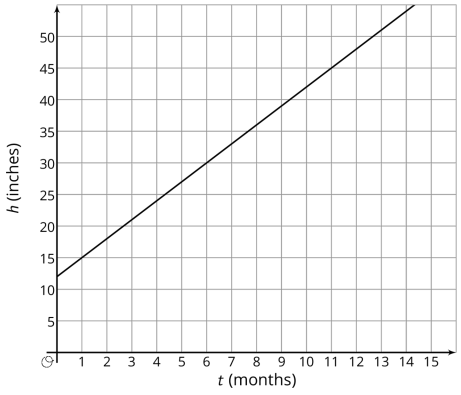
6. Customers at the gym pay a membership fee to join and then a fee for each class they attend. Here is a graph that represents the situation.

a. What does the slope of the line shown by the points mean in this situation?

b. What does the vertical intercept mean in this situation?



7. The graph shows the height in inches, h, of a bamboo plant t months after it has been planted.



a. Write an equation that describes the relationship between h and t.

b. After how many months will the bamboo plant be 66 inches tall? Explain or show your reasoning.

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